

BAKER, MT, USA

WMO: 726777

Lat: 46.346N

Lon: 104.258W

Elev: 906

StdP: 90.91

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94055

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-22.8	-28.9	0.3	-25.7	-26.0	0.4	-22.8	14.7	-6.3	13.5	-5.2	3.6	280	0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	35.0	18.4	32.9	18.2	31.2	17.8	21.2	30.2	20.1	28.9	19.0	27.7	5.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.6	24.6	17.3	13.8	23.4	16.2	12.9	22.3	65.8	30.3	61.3	28.8	57.7	27.6	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.8	10.7	-31.0	38.4	2.7	2.0	-33.0	39.9	-34.5	41.1	-36.0	42.2	-38.0	43.7		
(5)			-31.3	23.3	2.6	2.1	-33.2	24.8	-34.7	26.0	-36.1	27.1	-38.0	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-7.0	-6.7	0.0	6.0	11.3	17.0	22.1	20.7	15.1	6.9	0.4	-6.3		
	DBStd	11.75	7.38	7.75	7.34	5.29	4.64	3.95	3.51	3.67	4.98	5.23	6.31	7.48		
	HDD10.0	2417	526	469	314	138	40	2	0	0	12	121	290	506		
	HDD18.3	4522	784	702	568	371	221	69	8	19	122	354	538	764		
	CDD10.0	1214	0	0	4	17	81	213	376	332	164	26	2	0		
	CDD18.3	276	0	0	0	1	3	30	125	93	24	1	0	0		
	CDH23.3	4114	0	0	2	19	88	454	1703	1338	475	34	1	0		
	CDH26.7	1856	0	0	0	4	17	152	841	632	202	8	0	0		
(14) Wind	WSAvg	4.9	5.0	4.6	5.2	5.6	5.4	5.0	4.6	4.5	4.5	4.8	4.7	4.7		
(15) Precipitation	PrecAvg	367	9	9	15	35	70	71	47	32	32	30	12	7		
	PrecMax	524	32	42	48	101	187	189	186	106	78	93	75	22		
	PrecMin	221	0	0	3	7	23	14	6	2	8	7	0	0		
	PrecStd	73	7	8	10	23	45	37	36	22	20	22	14	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	13.8	22.1	26.1	28.9	33.7	38.1	37.4	34.1	27.3	20.2	12.2		
		MCWB	4.5	6.6	10.1	12.0	15.9	18.5	19.8	19.3	17.3	14.0	9.7	5.8		
	2%	DB	6.8	8.7	17.5	21.6	25.9	30.8	35.7	34.5	31.9	22.8	16.0	8.0		
		MCWB	2.7	3.9	8.3	10.9	15.1	18.5	18.9	18.2	16.6	11.7	7.6	3.1		
	5%	DB	4.5	5.8	13.9	18.2	23.4	27.9	33.6	32.4	28.9	19.0	12.5	5.6		
		MCWB	1.4	2.1	6.3	9.3	14.1	17.7	18.7	18.0	15.5	10.3	6.0	1.6		
	10%	DB	2.8	3.0	10.5	16.0	21.1	26.0	31.5	30.5	25.8	16.2	9.6	3.0		
		MCWB	0.3	0.5	5.1	8.1	12.9	16.4	18.4	17.6	14.5	9.0	4.4	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	7.1	10.6	13.3	18.5	21.9	23.4	22.6	19.3	15.0	10.2	5.9		
		MCDB	9.7	13.2	20.9	23.9	25.0	28.7	31.4	31.7	29.2	25.2	19.1	11.5		
	2%	WB	2.9	4.1	8.7	11.4	16.5	20.1	21.6	20.6	17.9	12.8	8.1	3.4		
		MCDB	6.1	8.3	16.8	19.9	23.4	27.4	30.8	30.5	27.6	20.5	15.4	7.4		
	5%	WB	1.7	2.2	6.9	9.9	15.0	18.8	20.4	19.5	16.8	11.0	6.2	1.9		
		MCDB	4.3	5.4	13.4	17.4	21.7	25.8	29.4	28.7	26.1	18.2	12.0	5.4		
	10%	WB	0.5	0.8	5.2	8.7	13.5	17.6	19.3	18.5	15.2	9.5	4.5	0.6		
		MCDB	2.7	3.1	10.2	15.0	19.8	24.2	28.2	27.2	24.0	15.5	9.2	3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.1	12.2	13.0	13.0	13.4	15.7	16.0	15.3	13.2	12.4	10.8		
		MCDBR	11.9	13.4	17.9	18.3	17.3	17.2	19.0	19.5	20.8	18.9	17.6	13.4		
		MCWBR	8.7	9.3	10.0	9.6	8.6	7.8	6.7	7.3	8.3	9.9	10.4	9.1		
		MCDBR	11.6	12.9	17.3	16.8	15.3	15.3	16.6	16.3	18.0	17.4	16.8	12.8		
	5% WB	MCWBR	8.7	9.2	9.9	9.2	8.1	7.7	7.0	7.3	7.6	9.3	10.2	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.256	0.273	0.307	0.328	0.341	0.353	0.355	0.320	0.288	0.259	0.246		
	taud		2.546	2.556	2.562	2.458	2.462	2.458	2.449	2.416	2.500	2.572	2.547	2.544		
	Ebn at Noon		883	945	970	953	937	922	906	891	900	890	861	849		
	Edh at Noon		60	73	85	103	107	108	108	107	90	72	60	54		
(44) All-Sky Solar Radiation	RadAvg		1.68	2.64	3.81	4.96	5.86	6.75	7.14	5.92	4.46	2.86	1.83	1.33		
	RadStd		0.13	0.19	0.29	0.27	0.57	0.46	0.27	0.34	0.32	0.25	0.19	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page